OE5450 (E) - NUMERICAL TECHNIQUES IN OCEAN HYDRODYNAMICS

Final Project

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Blood flow through arteries using the FVM analysis

Objective and Scope:

The primary objective of this report is to provide a thorough analysis of blood flow through arteries utilizing the Finite Volume Method. This investigation seeks to deepen our understanding of how various geometric and physiological factors impact crucial hemodynamic parameters. The report specifically focuses on key areas, including:

The sensitivity of coronary hemodynamic to the structural characteristics of blood vessels encompasses factors such as bifurcation angles, positional relationships, diameter ratios, and the tortuosity of arteries. The influence of stenotic lesions—narrowing in the arterial path—on flow patterns, pressure distributions, and wall shear stress is vital for understanding cardiovascular health. Implementing non-Newtonian fluid models is essential for accurately representing the complex rheological behaviour of blood, which is essential for realistic simulations.

The application of non-Newtonian fluid models to capture the complex rheological behaviour of blood. This report contributes to the growing body of knowledge on arterial hemodynamic and its implications for cardiovascular health and disease by synthesizing theoretical frameworks with computational simulations and experimental validations. The findings have potential applications in predicting disease progression, designing medical interventions, and developing personalized treatment strategies based on patient-specific vascular geometries.